REQUIREMENT ANALYSIS

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Introduction

The Script-Controlled ACL project is designed to provide secure and flexible access control for records based on specific field values and user conditions. The system uses a custom ACL script to evaluate access requirements before allowing a user to view or access a record.

2. Functional Requirements

- The system shall authenticate users before providing access to records.
- The system shall identify the current user's role and relevant user conditions.
- The system shall check the specified field values of a record.
- The system shall execute the configured ACL script when access is requested.
- The system shall grant access only when the ACL script evaluates to true.
- The system shall deny access when the required conditions are not satisfied.
- The system shall allow administrators to define or modify access conditions.
- The system shall apply the access-control rules consistently to the selected records.

3. Non-Functional Requirements

- **Security:** The system must prevent unauthorized users from accessing restricted records.
- **Performance:** Access decisions should be processed within a reasonable response time.
- **Reliability:** ACL rules should consistently provide the expected access decision.
- **Maintainability:** The access-control scripts should be easy to update and maintain.
- **Scalability:** The system should support additional records, users, and access conditions.
- **Usability:** Access rules should be understandable and manageable by authorized administrators.

4. Hardware Requirements

- **Processor:** Intel Core i3 or above
- **RAM:** Minimum 4 GB
- **Storage:** Minimum 10 GB free space
- **Network:** Internet connection for system access and configuration

5. Software Requirements

- **Operating System:** Windows / Linux / macOS
- **Web Browser:** Google Chrome, Microsoft Edge, or Mozilla Firefox
- **Development Environment:** Visual Studio Code or equivalent IDE
- **Scripting Support:** JavaScript
- **Access Control:** ACL-based authorization mechanism
- **Database:** A database system to store users, records, and field values

6. User Requirements

- The user must be authenticated before accessing protected records.
- The user must have the required permissions to request access.
- The user must satisfy the conditions defined by the ACL script.
- Administrators must be able to configure and manage access-control conditions.

7. Security Requirements

The system must validate every access request against the configured ACL conditions. Sensitive records must not be exposed when the required field or user conditions are not satisfied. The system should ensure that unauthorized users cannot bypass the ACL rules.

8. Expected Requirement Outcome

The requirement analysis defines the functional, non-functional, hardware, software, user, and security requirements needed to implement the Script-Controlled ACL system. These requirements provide a clear foundation for developing a secure record-level access-control mechanism based on field values and user conditions.